ETALE EXTENSIONS OF A DEDEKIND DOMAIN ARE FINITE PRODUCTS OF DEDEKIND DOMAINS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Etale maps preserve the dimension of localisations at corresponding primes and preserve regularity. Lemma \ref{lemma-dimension-etale-extension} and Lemma \ref{lemma-regular-etale-extension}, cited at proof lines 10889-10891. Identify the localisations at maximal ideals as regular local rings of dimension one.
- Characterisation of discrete valuation rings, and of Dedekind domains. Algebra, Lemma \ref{algebra-lemma-characterize-dvr} and Algebra, Lemma \ref{algebra-lemma-characterize-Dedekind}, cited at proof lines 10891 and 10896-10897. Turn the local descriptions into the stated global structure.
- A Noetherian normal ring is a finite product of normal domains. Algebra, Lemma \ref{algebra-lemma-characterize-reduced-ring-normal}, cited at proof lines 10893-10895. Gives the product decomposition.

Source reference. Original lemma 0AP2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13460–13460. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07QP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9839–9839. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AH0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13213–13213. Navigation pointer to a statement the

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Source reference. Original lemma 00PD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 927–927. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 030C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3943–3943. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 034X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4108–4108. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *If $A \to B$ is an étale ring map and A is a Dedekind domain, then B is a finite product of Dedekind domains. In particular, the localizations $B_{\mathfrak{q}}$ for $\mathfrak{q} \subset B$ maximal are discrete valuation rings.*

Proof. The statement on the local rings follows from Lemmas 07QP and 0AH0 and Algebra, Lemma 00PD. It follows that B is a Noetherian normal ring of dimension 1. By Algebra, Lemma 030C we conclude that B is a finite product of normal domains of dimension 1. These are Dedekind domains by Algebra, Lemma 034X. □